Emir Muhammet Aran

Computer Engineering Student | Medical AI Developer

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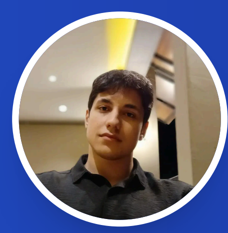

Education

Gazi University — 2025 - 2027
B.S. in Computer Engineering
Third Year - GPA 3.52 - Currently Enrolled

Ankara Medipol University — 2022 - 2025
B.S. in Computer Engineering
Sophomore: GPA 3.70 | Freshman: GPA 3.68 (Ranked 2nd)

Work Experience

Software Developer Intern — Summer 2024
Gibrin R&D | TOBB Garaj

- Developed cross-platform mobile applications using Flutter/Dart framework
- Built real-time chat application with Firebase integration (3300+ lines of code)
- Designed IoT monitoring system integrating ESP32 hardware with cloud database
- Implemented secure authentication, role-based access control, and data encryption
- Created weather application with API integration, GPS services, and custom animations

Medical AI Projects

BraTS 2020 Brain Tumor Segmentation

3D medical image segmentation using U-Net with EfficientNetB0 achieving 90.34% Sensitivity and 99.96% Specificity for precise tumor region identification (Necrotic, Enhancing Tumor, Edema). Implements Focal Loss for class imbalance. Interactive Gradio interface with dual-mode inference and ground truth comparison.

U-Net | 90.34% Sensitivity | 3D Segmentation | Focal Loss | ✓ DEPLOYED

Live Demo GitHub →

Brain Tumor MRI Classification System

Multi-class deep learning classifier for brain tumor detection (Glioma, Meningioma, Pituitary, No Tumor). Transfer learning with EfficientNetB3 achieving 99% accuracy with Grad-CAM visualizations for clinical interpretability.

EfficientNetB3 | 99% Acc | Transfer | Grad-CAM | ✓ DEPLOYED

Live Demo GitHub →

Breast Cancer Histopathology Detection

Deep learning system for metastatic cancer detection in histopathology images achieving 91% sensitivity (exceeds FDA screening benchmark of 90%) and AUC 0.94. Comprehensive documentation including FDA compliance considerations, ethical guidelines, and clinical validation framework.

91% Sensitivity | AUC 0.94 | CNN | Focal Loss | ✓ DEPLOYED

Live Demo GitHub →

Chest X-ray Disease Classification

Multi-label deep learning system for 15-class thoracic disease detection from chest X-ray images using EfficientNetB0 with full fine-tuning. Achieves 0.784 mean AUC with Test-Time Augmentation (TTA). Implements Focal Loss for class imbalance, balanced sampling with oversampling for rare diseases, and medical-appropriate threshold optimization for ≥80% recall priority.

EfficientNetB0 | 0.784 AUC | Multi-Label (15 diseases) | Focal Loss | TTA | ✓ OPEN SOURCE

Live Demo GitHub →

Additional Technical Projects

Computer Vision
YOLO/Faster R-CNN object detection, face recognition (OpenCV, Dlib, MTCNN), handwritten digit/character recognition (MNIST/EMNIST), transfer learning with ResNet/VGG16/MobileNet

Natural Language Processing
RNN/LSTM sequence models, transformer-based automatic subtitle translation with attention mechanisms, text classification systems

Predictive Analytics
Time series forecasting (FBProphet for Bitcoin), ensemble methods (Random Forest, XGBoost), regression models (linear, polynomial, logistic) for various prediction tasks

Mobile & IoT Development
Cross-platform Flutter applications, real-time Firebase integration, IoT sensor systems with ESP32/Arduino, cloud database connectivity

View all projects on GitHub (20+ repositories)

Technical Skills

Programming Languages
Python, C, C#, Java, Dart, LEGv8 Assembly, Bash

AI/ML Frameworks
PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, XGBoost

Data Science & Analysis
Pandas, NumPy, Matplotlib, Feature Engineering, Statistical Analysis

Development & Deployment
Flutter, Firebase, HuggingFace Spaces, REST APIs, Git, Linux

Hardware Integration
Arduino, ESP32, IoT Systems, Sensor Integration

Game Development
Unity, Unreal Engine, Rapid Prototyping

Achievements & Leadership

- 🏆 **Ostim Game Jam 2025** - 1st Place Winner | Led team in rapid game prototype development
- 🥈 **Pura Game Jam 2025** - 2nd Place Winner | Collaborative problem-solving under tight deadlines
- 🏅 **Ankü Game Jam 2025** - 5th Place | Demonstrated creativity and technical execution
- 🏅 **Ayaz Jam 2025** - 5th Place | Demonstrated teamwork and creativity
- 👥 **MEDCODES** - Board Member (2024-Present) | Presented Unity workflow training sessions
- 👥 **CYBERMEDU** - Board Member (2023-Present) | Contributing to cybersecurity education initiatives

Languages

English: Advanced (C1)
Turkish: Native